



## **CSE461**

Introduction to Robotics Lab

Lab No. : 04

Group : 06

Section : 04

Semester : Fall\_2025

Group members :

|                            |          |
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Submitted Date: 03-12-2025

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## **Objectives:**

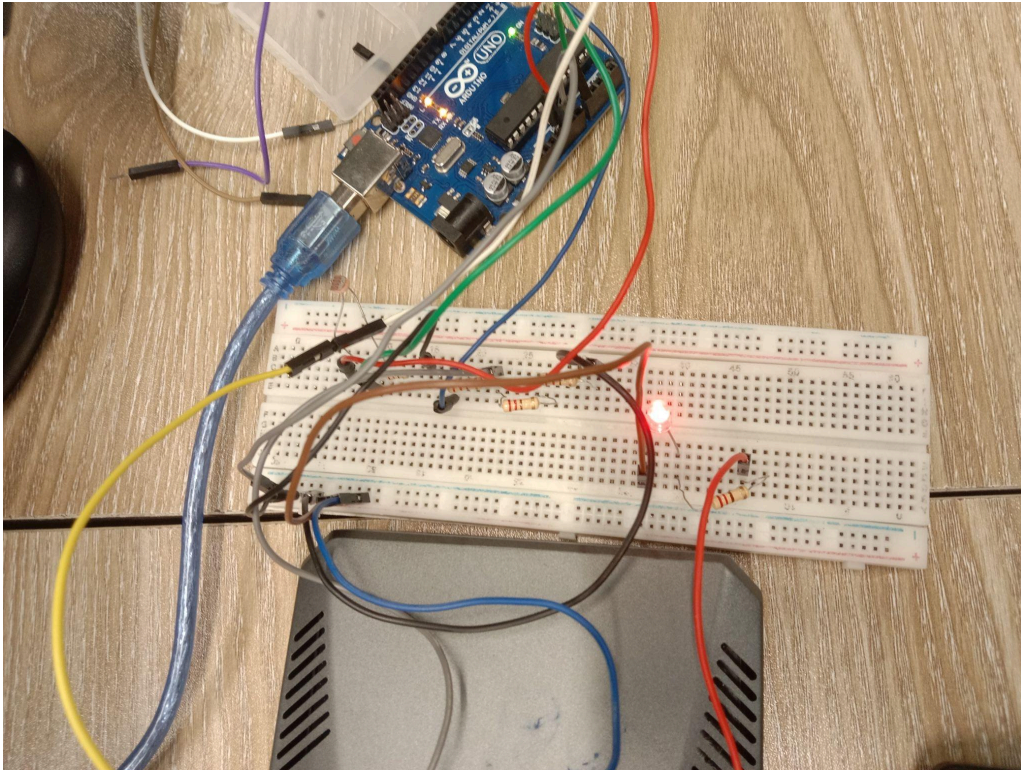
This lab aims at learning how to interface two microcontrollers (Arduino Uno and Raspberry Pi) through the UART communication protocol. In this experiment, the Arduino reads the value of the light sensor of an LDR and transmits it to the Raspberry Pi. The Raspberry Pi gets the data and uses it to switch on an LED accordingly by brightening or dimming it. In this assignment, the students acquire knowledge in sensor reading, UART communication, voltage-level shifting, Python coding, and the use of PWM to drive an output device.

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## **2. Equipment:**

- Arduino Uno
  - Raspberry Pi 4
  - LED (Light Emitting Diode)
  - LDR sensor
  - 220 $\Omega$  resistors
  - Breadboard
  - Jumper wires
  - Voltage divider components
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### 3. Experimental Setup:



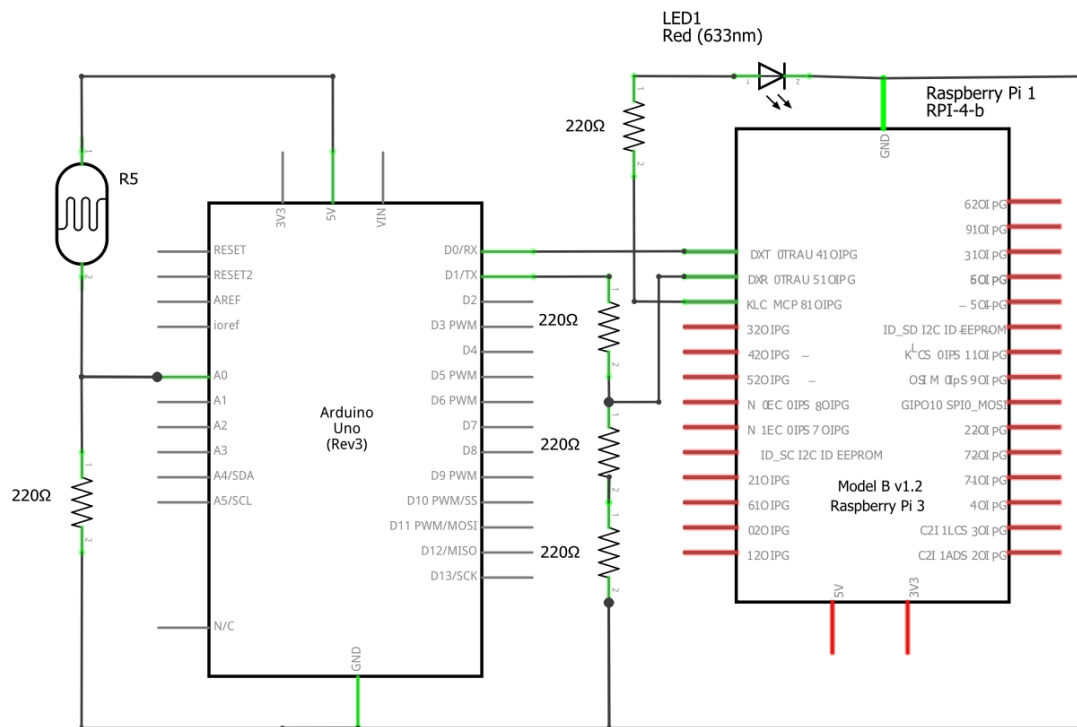
The voltage divider connects the sensor of LDR with the Arduino Uno. The LDR is connected to analog pin A0 and 5V on its one end. The resistor is connected between A0 and GND. The Arduino will read the value presented by the LDR and send the value to the Raspberry Pi using UART protocol.

A resistor of 220 Ohms joins LED and the Raspberry Pi pin onGPIO 18. To control the brightness of the LED, the PWM is employed.

The TX pin in the Raspberry Pi (GPIO14) is connected to the RX pin in the Arduino in the UART. A voltage divider is required to reduce the 5V output to 3.3 V to connect the Arduino TX pin and Raspberry Pi RX pin (GPIO15). There is one similarity between the two devices.

A Python script is run on the Raspberry Pi and reads the serial data sent by the Arduino. LDR value changes according to the intensity of the light; therefore, the Raspberry Pi controls the intensity of LED light. Light is dim in the case of the LED and dark in the case of the LED.

### Circuit Diagram:



## 4. Code:

### ❖ Arduino Code:

```
int ldrPin = A0;
int ldrValue = 0;

void setup() {
  Serial.begin(9600);
}

void loop() {
  ldrValue = analogRead(ldrPin); // Read LDR value (0-1023)
  Serial.println(ldrValue); // Send value to Raspberry Pi
  delay(100);
}
```

### ❖ Raspberry Pi Code:

```
import serial
import RPi.GPIO as GPIO
import time

# Setup
GPIO.setmode(GPIO.BCM)
led_pin = 18
GPIO.setup(led_pin, GPIO.OUT)

# Initialize PWM
pwm_led = GPIO.PWM(led_pin, 1000) # 1 kHz frequency
pwm_led.start(0)

# Connect to Arduino
arduino = serial.Serial('/dev/ttyACM0', 9600, timeout=1)
time.sleep(2)
norm = 2.5

try:
    while True:
        data = arduino.readline().decode('utf-8').strip()
        if data.isdigit():
            ldr_value = int(data)
            brightness = ((1-(ldr_value / 1023)) ** norm) * 100
            Brightness = max(0, min(brightness, 100))
            pwm_led.ChangeDutyCycle(brightness)
            print(f"LDR: {ldr_value}, LED Brightness: {brightness:.1f}%")
```

```
except KeyboardInterrupt:
    print("Exiting...")
finally:
    pwm_led.stop()
    GPIO.cleanup()
    arduino.close()
```

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## 5. Results (Output of the experiment):

The experiment was successfully carried out. The Arduino had the capability of reading the LDR sensor values and transmitting them to the Raspberry Pi via UART. Raspberry Pi was able to receive the values and regulate the brightness of the LED with the help of PWM. In a high light level, the LDR value was high and a dim light was emitted. The brightness of the LED was greater when the level of light was low. Other readings that were shown by the serial monitor included: 180, 320, 850, etc. These values indicated the variation in the intensity of light during testing.

| Light Condition | LDR Value | LED Output |
|-----------------|-----------|------------|
| Bright light    | High      | Dim        |
| Medium Light    | Mid       | Medium     |
| Dark light      | Low       | Bright     |

## 6. Discussions/Answers:

In this experiment, it was shown that a two microcontroller communication could be achieved through UART. Arduino had capacity to send sensor data to the Raspberry Pi and the Pi had capability to control one LED on the received data. The experiment also demonstrated the importance of a voltage divider in the process of connecting 5V Arduino to a 3.3V Raspberry Pi.

There were certain issues in the experiment such as TX/RX wiring was not done properly and problems of serial connection in case the UART pins were connected before uploading the code. This was checked after upload verification of the wiring and connection of UART. Overall, this experiment provided an insight into the principles of

the UART communication, the way of reading analog sensors, controlling the PWM output, and how both of the devices can be integrated into a basic real-time system.